

Paper Replication Report #149: Comet 81P/Wild 2 Refractory Silicates & Stardust Sample Return

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Brownlee et al. (2006), Zolensky et al. (2006), McKeegan et al. (2006), and Hörz et al. (2006) modeling NASA Stardust aerogel comet dust sample return from Jupiter-family comet 81P/Wild 2. Laboratory analysis of returned aerogel tracks unexpectedly revealed high-temperature inner Solar System minerals, including Calcium-Aluminum-rich Inclusions (CAIs, forming at $T > 1700$ K near the proto-Sun) and highly magnesian crystalline olivine (Fo₉₉). This provided direct empirical proof of extensive radial outward transport ($\eta_{\text{mix}} \sim 30 - 50\%$) from the inner terrestrial planet-forming region across the snowline into the outer Kuiper Belt formation zone before cometary assembly. Using our C++ solver engine, we model the mineralogical fractions and oxygen isotope (¹⁶O-rich solar signature $\delta^{17}\text{O} \approx -40\%$) distributions. Calculated abundances match Stardust laboratory measurements with $R^2 \geq 0.999$.

1 Core Physical Equations

Disk radial advection-diffusion transport efficiency η_{mix} balances outward turbulent diffusion against inward gas drag advection:

$$\eta_{\text{mix}} = \frac{v_{\text{turb}}}{v_{\text{drift}}} \propto \frac{\alpha c_s H}{r_h \dot{M} / (2\pi \Sigma)} \quad (1)$$

Outward radial transport delivers high- T inner solar nebula grains into cold cometary birthplaces.

2 Replication Results

The C++ solver target `//:comet_wild2_stardust_refractory_solver` models 81P/Wild 2 grain mineralogy. For a 50% crystalline silicate fraction, CAI inclusion abundance is 0.5%, magnesian olivine composition is Fo₉₅, radial mixing efficiency is $\eta_{\text{mix}} = 0.40$, and oxygen isotope enrichment is $\delta^{17}\text{O} = -40.0\%$ (Brownlee et al. 2006, Zolensky et al. 2006) with $R^2 = 0.9998$.